How Different Uses of AI Shape Labor Demand: Evidence from France

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Data and replication programs

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Overview

The code in this replication package constructs the analysis file from the three data sources (BTS Matched employer-employee data; BIC-IS balance sheet data; Survey on information and communication technologies in business) using SAS, R and Stata. The main file run all of the code to generate the data for the figures and tables in the paper. The replicator should expect the code to run for approximately 6 to 8 hours.

1. Data Availability

I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript. Some data cannot be made publicly available.
Access to some confidential data, on which is based this work, has been made possible within a secure environment offered by CASD – Secured Remote Access Device (Ref. 10.34724/CASD). Any researcher can access the same datasets following the instructions provided at https://www.casd.eu/en/.

2. Details on Each Data Source

The main datasets are listed on this webpage (https://www.casd.eu/en/data-used-at-casd/):

- BTS (Matched employer-employee data): These files describe the job, employee individual characteristics and more information on the employer from the Register of enterprises and establishments. (https://www.casd.eu/en/source/all-employees-databases-job-position-data/)
- BIC-IS: These files contain information about firm level tax reports (balance sheet and profit and losses statement) provided by the French fiscal administration. (https://www.casd.eu/en/source/industrial-and-commercial-profits-normal-scheme/)
- Survey on information and communication technologies in business: The EC survey on information and communication technologies (ICT) use and electronic commerce in businesses is part of the European survey program, in accordance with the Commission implementing regulation (EU) FRIBS 2020/1197 of July 30, 2020. This program serves to evaluate the progress of use of ICTs by European businesses. The aim of the ICT survey is to improve understanding of computerisation and the dissemination of information and communication technologies in business. It aims notably to assess the place of new tools in external business relations (internet, electronic commerce) and in their internal working (networks, integrated management systems). See https://www.casd.eu/en/source/information-and-communication-technologies-survey-businesses-section/

In addition to confidential dataset, some publicly available data are joined:

- data\source\bergeaud_2024_exposure_ai_final.dta contains measures of the exposure and substitutability of French occupations to AI, based on Bergeaud

(2024)[1], who adapts the methodologies of Gmyrek et al. (2023)[2] and Pizzinelli et al. (2023)[3] to the French context.
- data\other\occupation_level_2digit.dta classifies occupations in low/middle/high skilled occupations.
- data\other\passage_naf1_naf2_maj.dta provides a table de passage between the version NACE rev. 1 classification and the version NACE rev. 2.

<u>3. Computational Requirements</u>

The code was last run using Stata version 17.0, R version 4.3.1, and SAS version 9.4. The estimated time required to reproduce the analyses on a standard CASD access (2025) is approximately 6 to 8 hours. The estimated storage space required ranges between 2 GB and 25 GB.

<u>4. Instructions to Replicators</u>

The replication programs are contained in the file named “code”. They require the datasets listed previously.

By running the "0_main.do" file, you will obtain the results presented in the paper (four figures and one table). Please ensure that you specify the file path at the beginning of the code, corresponding to the location where you downloaded this replication file. The scripts are organized as follows:

- Files with the "1_" prefix clean the datasets.
- Files with the "2_" prefix merge all the cleaned datasets.
- Files with the "3_" prefix generate descriptive statistics.
- Files with the "4_" prefix produce the estimation results.
- Files with the "5_" prefix delete the remaining datasets.

[1] Bergeaud, A. (2024), « Exposition à l’intelligence artificielle générative et emploi : une application à la classification socio-professionnelle française », *mimeo.*
[2] Gmyrek, P., Berg, J., & Bescond, D. (2023). « Generative AI and Jobs: A global analysis of potential effects on job quantity and quality ». *ILO Working Paper, 96*.
[3] Pizzinelli, C., Panton, A., Mendes Tavares, M., Cazzaniga, M., & Li, L. (2023). « Labor Market Exposure to AI: Cross-country Differences and Distributional Implications. », *IMF Working Paper*.